

# Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two\_stage - IPDE alpha = 0.6 - both utilities - N = 30

|                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|--------------------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 1                           |        |        |        |        |        |        |
| DLT rate                             | 0.3    | 0.35   | 0.4    | 0.45   | 0.5    |        |
| Efficacy rate                        | 0.05   | 0.06   | 0.07   | 0.08   | 0.09   |        |
| Utility 2                            | -0.2   | -0.3   | -0.3   | -0.4   | -0.4   |        |
| Utility 3                            | 31     | 29.6   | 28.2   | 26.8   | 25.4   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 32.6   | 8.7    | 1.5    | 0.1    | 0.2    | 56.9   |
| U3 - Toxicity prior Beta(0.15, 0.85) | 35.7   | 7      | 0.7    | 0.2    | 0      | 56.4   |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 34.2   | 9.3    | 1.2    | 0.9    | 0.2    | 54.2   |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 37.2   | 6.8    | 0.7    | 0.5    | 0.1    | 54.7   |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 34     | 9.8    | 1.6    | 1      | 0.3    | 53.3   |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 37.7   | 7.2    | 1      | 0.3    | 0.2    | 53.6   |
| U2 - Toxicity prior Beta(1, 1)       | 33     | 9.6    | 2.1    | 0.7    | 0.4    | 54.2   |
| U3 - Toxicity prior Beta(1, 1)       | 36.9   | 7.4    | 1.1    | 0.2    | 0.1    | 54.3   |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 15.1   | 4.2    | 2.2    | 0.9    | 0.4    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 15.1   | 4.3    | 2.2    | 0.8    | 0.5    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 14.6   | 4.2    | 2.5    | 1.2    | 0.5    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 14.6   | 4.3    | 2.6    | 1.1    | 0.5    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 14.4   | 4.1    | 2.8    | 1.3    | 0.6    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 14.5   | 4      | 2.7    | 1.3    | 0.6    |        |
| U2 - Toxicity prior Beta(1, 1)       | 14.4   | 4      | 2.6    | 1.4    | 0.7    |        |
| U3 - Toxicity prior Beta(1, 1)       | 14.3   | 4      | 2.7    | 1.4    | 0.7    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.7    | 2      | 0.5    | 0.3    | 0.1    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.6    | 2      | 0.5    | 0.3    | 0.1    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.6    | 1.8    | 0.5    | 0.3    | 0.1    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.6    | 1.9    | 0.5    | 0.3    | 0.1    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.5    | 1.8    | 0.4    | 0.3    | 0.1    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.5    | 1.8    | 0.4    | 0.3    | 0.1    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.4    | 1.5    | 0.4    | 0.3    | 0      |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.4    | 1.5    | 0.4    | 0.3    | 0      |        |
| Scenario 2                           |        |        |        |        |        |        |
| DLT rate                             | 0.15   | 0.2    | 0.3    | 0.35   | 0.45   |        |
| Efficacy rate                        | 0.2    | 0.25   | 0.5    | 0.5    | 0.5    |        |
| Utility 2                            | 0.1    | 0.1    | 0.2    | 0.1    | 0.1    |        |
| Utility 3                            | 46     | 47     | 58     | 56     | 52     |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 36.3   | 17     | 31.3   | 10.5   | 3      | 1.9    |
| U3 - Toxicity prior Beta(0.15, 0.85) | 38.9   | 18.9   | 28.7   | 9.6    | 2      | 1.9    |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 34.5   | 16.1   | 31.7   | 13     | 2.8    | 1.9    |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 37.1   | 18.4   | 28.7   | 11.5   | 2.5    | 1.8    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 34.5   | 15.4   | 31.1   | 13.5   | 3.6    | 1.9    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 36.3   | 17.6   | 29     | 12.4   | 2.8    | 1.9    |
| U2 - Toxicity prior Beta(1, 1)       | 34.3   | 15.1   | 31.9   | 13     | 3.8    | 1.9    |
| U3 - Toxicity prior Beta(1, 1)       | 36.7   | 16.6   | 29.2   | 12.7   | 2.9    | 1.9    |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 11.5   | 5.7    | 6.8    | 3.7    | 2      |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 11.6   | 5.7    | 6.7    | 3.8    | 2      |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 10.9   | 5.2    | 6.9    | 4.4    | 2.3    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 10.9   | 5.2    | 6.8    | 4.5    | 2.3    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 10.7   | 5      | 6.7    | 4.8    | 2.5    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 10.7   | 5      | 6.8    | 4.8    | 2.5    |        |
| U2 - Toxicity prior Beta(1, 1)       | 10.7   | 4.9    | 6.6    | 4.9    | 2.6    |        |
| U3 - Toxicity prior Beta(1, 1)       | 10.7   | 4.9    | 6.6    | 4.9    | 2.6    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.5    | 2.5    | 1.3    | 0.8    | 0.3    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.5    | 2.5    | 1.3    | 0.8    | 0.3    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.4    | 2.3    | 1.2    | 0.9    | 0.3    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.4    | 2.3    | 1.2    | 0.9    | 0.3    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.3    | 2.2    | 1.2    | 0.9    | 0.3    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.4    | 2.2    | 1.2    | 0.9    | 0.3    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.4    | 2.2    | 1.1    | 0.8    | 0.3    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.4    | 2.2    | 1.1    | 0.9    | 0.3    |        |

# Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two\_stage - IPDE alpha = 0.6 - both utilities - N = 30

|                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|--------------------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 3                           |        |        |        |        |        |        |
| DLT rate                             | 0.15   | 0.2    | 0.3    | 0.35   | 0.45   |        |
| Efficacy rate                        | 0.1    | 0.4    | 0.4    | 0.4    | 0.4    |        |
| Utility 2                            | -0.1   | 0.2    | 0.1    | 0.1    | -0.1   |        |
| Utility 3                            | 40     | 56     | 52     | 50     | 46     |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 19     | 39.5   | 21.6   | 8.4    | 1.5    | 10     |
| U3 - Toxicity prior Beta(0.15, 0.85) | 20.3   | 42.7   | 18.7   | 7      | 1.1    | 10.2   |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 19.4   | 36.2   | 23.5   | 9.4    | 1.9    | 9.6    |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 20.8   | 40.6   | 20.2   | 7.6    | 1.2    | 9.6    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 18.7   | 35.3   | 23.8   | 9.8    | 2.9    | 9.5    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 20.4   | 39.3   | 20.3   | 8.8    | 1.7    | 9.5    |
| U2 - Toxicity prior Beta(1, 1)       | 18.9   | 35.2   | 23.1   | 10.6   | 2.7    | 9.5    |
| U3 - Toxicity prior Beta(1, 1)       | 21.7   | 38.8   | 18.7   | 9.2    | 2.1    | 9.5    |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 9.6    | 7.7    | 6.1    | 3.6    | 1.9    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 9.6    | 7.8    | 6      | 3.6    | 1.9    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 9.4    | 7      | 6.3    | 4.1    | 2.2    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 9.3    | 7      | 6.3    | 4.2    | 2.2    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 9.1    | 6.8    | 6.2    | 4.5    | 2.4    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 9.1    | 6.8    | 6.1    | 4.6    | 2.4    |        |
| U2 - Toxicity prior Beta(1, 1)       | 9      | 6.9    | 5.8    | 4.7    | 2.5    |        |
| U3 - Toxicity prior Beta(1, 1)       | 9      | 6.9    | 5.7    | 4.7    | 2.6    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.2    | 2.9    | 1.1    | 1      | 0.3    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.2    | 2.9    | 1.2    | 1      | 0.3    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.2    | 2.8    | 1.2    | 1      | 0.3    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.2    | 2.7    | 1.2    | 1.1    | 0.3    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.2    | 2.7    | 1.2    | 1      | 0.3    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.2    | 2.7    | 1.2    | 1      | 0.3    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.2    | 2.6    | 1.1    | 1      | 0.3    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.2    | 2.6    | 1.1    | 1      | 0.3    |        |
| Scenario 4                           |        |        |        |        |        |        |
| DLT rate                             | 0.05   | 0.1    | 0.18   | 0.3    | 0.4    |        |
| Efficacy rate                        | 0.05   | 0.1    | 0.25   | 0.5    | 0.55   |        |
| Utility 2                            | 0      | 0      | 0.1    | 0.2    | 0.1    |        |
| Utility 3                            | 41     | 42     | 47.8   | 58     | 57     |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 8.9    | 8.4    | 23.6   | 41.1   | 16.1   | 1.9    |
| U3 - Toxicity prior Beta(0.15, 0.85) | 14.1   | 10.8   | 19.9   | 39.8   | 13.5   | 1.9    |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 8.9    | 7.2    | 20.4   | 44.9   | 16.8   | 1.8    |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 12.9   | 8.8    | 19.4   | 43.1   | 14     | 1.8    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 8.1    | 6.9    | 19.5   | 44.6   | 19     | 1.9    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 13     | 9.2    | 16.8   | 44.4   | 14.8   | 1.8    |
| U2 - Toxicity prior Beta(1, 1)       | 8.6    | 6.7    | 17.7   | 45     | 20.1   | 1.9    |
| U3 - Toxicity prior Beta(1, 1)       | 10.8   | 9.3    | 16.4   | 44.4   | 17.3   | 1.8    |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 4.5    | 4.5    | 6.4    | 8.2    | 6.3    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 4.4    | 4.5    | 6.4    | 8.1    | 6.3    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 4.2    | 4.1    | 6.1    | 8.7    | 6.7    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 4.3    | 4.1    | 6.2    | 8.6    | 6.6    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 4.3    | 4      | 5.9    | 8.8    | 6.9    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 4.2    | 4      | 5.9    | 8.7    | 7      |        |
| U2 - Toxicity prior Beta(1, 1)       | 4.2    | 3.8    | 5.7    | 8.6    | 7.4    |        |
| U3 - Toxicity prior Beta(1, 1)       | 4.2    | 3.8    | 5.7    | 8.6    | 7.4    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.3    | 2.8    | 1.4    | 2.3    | 0.8    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.3    | 2.8    | 1.4    | 2.3    | 0.8    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.2    | 2.7    | 1.4    | 2.3    | 0.8    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.2    | 2.7    | 1.4    | 2.3    | 0.8    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.2    | 2.7    | 1.4    | 2.2    | 0.7    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.2    | 2.7    | 1.3    | 2.2    | 0.7    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.2    | 2.6    | 1.3    | 2.2    | 0.7    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.2    | 2.6    | 1.3    | 2.2    | 0.7    |        |

# Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two\_stage - IPDE alpha = 0.6 - both utilities - N = 30

|                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|--------------------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 5                           |        |        |        |        |        |        |
| DLT rate                             | 0.07   | 0.12   | 0.17   | 0.22   | 0.3    |        |
| Efficacy rate                        | 0.05   | 0.3    | 0.3    | 0.3    | 0.3    |        |
| Utility 2                            | 0      | 0.2    | 0.1    | 0.1    | 0      |        |
| Utility 3                            | 40.2   | 53.2   | 51.2   | 49.2   | 46     |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 5.1    | 31.1   | 29.2   | 20.3   | 10.1   | 4.2    |
| U3 - Toxicity prior Beta(0.15, 0.85) | 9.5    | 36.8   | 25.1   | 18.8   | 5.6    | 4.2    |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 5.7    | 30.3   | 27.4   | 21.5   | 10.9   | 4.2    |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 9      | 35.5   | 25.3   | 19.4   | 6.6    | 4.2    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 5.1    | 29.7   | 26.7   | 22.5   | 11.8   | 4.2    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 9      | 33.4   | 25.8   | 20.9   | 6.7    | 4.2    |
| U2 - Toxicity prior Beta(1, 1)       | 5      | 28.8   | 25.6   | 24.1   | 12.3   | 4.2    |
| U3 - Toxicity prior Beta(1, 1)       | 8      | 35.2   | 24.3   | 20.9   | 7.4    | 4.2    |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 4.9    | 6.3    | 6.7    | 6      | 5.7    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 4.9    | 6.3    | 6.6    | 6      | 5.7    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 4.8    | 5.9    | 6.6    | 6.2    | 6      |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 4.8    | 6      | 6.5    | 6.3    | 6      |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 4.8    | 5.9    | 6.4    | 6.3    | 6.2    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 4.7    | 5.9    | 6.3    | 6.4    | 6.2    |        |
| U2 - Toxicity prior Beta(1, 1)       | 4.7    | 5.8    | 6.2    | 6.3    | 6.4    |        |
| U3 - Toxicity prior Beta(1, 1)       | 4.7    | 5.7    | 6.3    | 6.2    | 6.5    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.2    | 3.2    | 1.7    | 2.2    | 1.1    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.2    | 3.3    | 1.7    | 2.2    | 1.1    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.2    | 3.2    | 1.7    | 2.2    | 1.1    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.2    | 3.2    | 1.7    | 2.2    | 1.1    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.2    | 3.2    | 1.6    | 2.2    | 1.1    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.1    | 3.2    | 1.7    | 2.2    | 1.1    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.1    | 3.2    | 1.7    | 2.1    | 1      |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.2    | 3.2    | 1.7    | 2.1    | 1      |        |
| Scenario 6                           |        |        |        |        |        |        |
| DLT rate                             | 0.29   | 0.34   | 0.38   | 0.43   | 0.52   |        |
| Efficacy rate                        | 0.3    | 0.32   | 0.35   | 0.38   | 0.4    |        |
| Utility 2                            | 0      | 0      | 0      | -0.1   | -0.1   |        |
| Utility 3                            | 46.4   | 45.6   | 45.8   | 45.6   | 43.2   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 68.3   | 13.3   | 2.6    | 1.4    | 0.4    | 14     |
| U3 - Toxicity prior Beta(0.15, 0.85) | 71.2   | 11.5   | 1.9    | 0.9    | 0.4    | 14.1   |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 67.9   | 13.2   | 3.6    | 1.3    | 0.4    | 13.6   |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 69.9   | 12.5   | 3      | 0.8    | 0.2    | 13.6   |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 66.6   | 13.9   | 3.7    | 1.7    | 0.7    | 13.4   |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 69.1   | 13.4   | 2.7    | 1.1    | 0.4    | 13.3   |
| U2 - Toxicity prior Beta(1, 1)       | 66.7   | 13.7   | 4.2    | 1.4    | 0.8    | 13.2   |
| U3 - Toxicity prior Beta(1, 1)       | 68.4   | 13.8   | 2.8    | 1.2    | 0.6    | 13.2   |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 19.3   | 4.3    | 2.2    | 1.1    | 0.5    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 19.2   | 4.4    | 2.2    | 1      | 0.5    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 18.6   | 4.4    | 2.6    | 1.3    | 0.6    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 18.6   | 4.4    | 2.6    | 1.3    | 0.6    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 18.4   | 4.4    | 2.6    | 1.5    | 0.7    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 18.4   | 4.4    | 2.6    | 1.5    | 0.7    |        |
| U2 - Toxicity prior Beta(1, 1)       | 18.3   | 4.6    | 2.6    | 1.4    | 0.7    |        |
| U3 - Toxicity prior Beta(1, 1)       | 18.3   | 4.5    | 2.6    | 1.4    | 0.7    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.4    | 1.6    | 0.5    | 0.3    | 0      |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.4    | 1.6    | 0.4    | 0.3    | 0.1    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.4    | 1.5    | 0.5    | 0.3    | 0.1    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.4    | 1.5    | 0.5    | 0.3    | 0.1    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.4    | 1.4    | 0.4    | 0.3    | 0.1    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.4    | 1.4    | 0.4    | 0.3    | 0.1    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.3    | 1.3    | 0.4    | 0.3    | 0      |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.3    | 1.3    | 0.4    | 0.3    | 0      |        |

# Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two\_stage - IPDE alpha = 0.6 - both utilities - N = 30

|                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|--------------------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 7                           |        |        |        |        |        |        |
| DLT rate                             | 0.16   | 0.2    | 0.26   | 0.31   | 0.36   |        |
| Efficacy rate                        | 0.2    | 0.5    | 0.5    | 0.5    | 0.5    |        |
| Utility 2                            | 0      | 0.3    | 0.2    | 0.2    | 0.1    |        |
| Utility 3                            | 45.6   | 62     | 59.6   | 57.6   | 55.6   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 26.5   | 36.2   | 21.8   | 9.6    | 3.4    | 2.5    |
| U3 - Toxicity prior Beta(0.15, 0.85) | 28.6   | 39.5   | 19.5   | 8      | 1.9    | 2.5    |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 27.1   | 33.1   | 21.1   | 12     | 4.4    | 2.3    |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 28     | 38.4   | 18.7   | 9.9    | 2.7    | 2.3    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 26     | 33.9   | 21.1   | 11.9   | 4.8    | 2.3    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 27.6   | 37     | 19.2   | 10.3   | 3.6    | 2.3    |
| U2 - Toxicity prior Beta(1, 1)       | 25.2   | 34.1   | 20.4   | 13.3   | 4.7    | 2.3    |
| U3 - Toxicity prior Beta(1, 1)       | 26.7   | 38.4   | 18.5   | 10.7   | 3.4    | 2.3    |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 10.7   | 7.3    | 5.6    | 3.5    | 2.4    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 10.7   | 7.2    | 5.7    | 3.6    | 2.4    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 10.3   | 6.6    | 5.6    | 4.3    | 2.8    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 10.2   | 6.7    | 5.6    | 4.3    | 2.8    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 10.1   | 6.6    | 5.6    | 4.4    | 3      |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 10.1   | 6.6    | 5.6    | 4.4    | 2.9    |        |
| U2 - Toxicity prior Beta(1, 1)       | 10     | 6.7    | 5.5    | 4.4    | 3.1    |        |
| U3 - Toxicity prior Beta(1, 1)       | 10     | 6.7    | 5.5    | 4.4    | 3.1    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.2    | 2.7    | 1      | 0.9    | 0.4    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.2    | 2.6    | 1.1    | 0.9    | 0.4    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.2    | 2.5    | 1.1    | 1      | 0.4    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.2    | 2.5    | 1      | 1      | 0.4    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.2    | 2.5    | 1      | 0.9    | 0.4    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.2    | 2.5    | 1      | 1      | 0.4    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.2    | 2.4    | 1      | 0.9    | 0.3    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.2    | 2.5    | 1      | 0.9    | 0.3    |        |
| Scenario 8                           |        |        |        |        |        |        |
| DLT rate                             | 0.32   | 0.37   | 0.42   | 0.48   | 0.52   |        |
| Efficacy rate                        | 0.1    | 0.15   | 0.2    | 0.3    | 0.4    |        |
| Utility 2                            | -0.2   | -0.2   | -0.2   | -0.2   | -0.1   |        |
| Utility 3                            | 33.2   | 34.2   | 35.2   | 38.8   | 43.2   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 44     | 8      | 2.9    | 1.3    | 0.3    | 43.5   |
| U3 - Toxicity prior Beta(0.15, 0.85) | 46.3   | 6.9    | 2.4    | 1.1    | 0.2    | 43.1   |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 44     | 9.1    | 3      | 1.3    | 0.7    | 41.9   |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 46.3   | 8      | 2.3    | 1      | 0.3    | 42.1   |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 44.2   | 8.8    | 2.8    | 1.8    | 0.6    | 41.8   |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 45.6   | 7.9    | 2.8    | 1.2    | 0.4    | 42.1   |
| U2 - Toxicity prior Beta(1, 1)       | 43.5   | 8.9    | 3.4    | 1.6    | 0.6    | 42     |
| U3 - Toxicity prior Beta(1, 1)       | 45.1   | 8.2    | 3.1    | 1.1    | 0.5    | 42     |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 17     | 4      | 2      | 0.8    | 0.3    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 17.1   | 3.8    | 2.1    | 0.8    | 0.3    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 16.6   | 4      | 2.4    | 0.9    | 0.4    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 16.6   | 4      | 2.3    | 0.9    | 0.4    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 16.5   | 3.7    | 2.5    | 1.1    | 0.4    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 16.5   | 3.7    | 2.5    | 1.1    | 0.5    |        |
| U2 - Toxicity prior Beta(1, 1)       | 16.4   | 3.9    | 2.4    | 1.1    | 0.5    |        |
| U3 - Toxicity prior Beta(1, 1)       | 16.4   | 3.9    | 2.4    | 1.1    | 0.5    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.4    | 1.6    | 0.4    | 0.2    | 0      |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.4    | 1.6    | 0.4    | 0.2    | 0      |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.4    | 1.5    | 0.4    | 0.2    | 0      |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.4    | 1.5    | 0.4    | 0.2    | 0      |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.3    | 1.4    | 0.3    | 0.2    | 0      |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.3    | 1.4    | 0.3    | 0.2    | 0      |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.3    | 1.2    | 0.3    | 0.2    | 0      |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.3    | 1.3    | 0.3    | 0.2    | 0      |        |



# Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two\_stage - IPDE alpha = 0.6 - both utilities - N = 30

|                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|--------------------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 9                           |        |        |        |        |        |        |
| DLT rate                             | 0.15   | 0.18   | 0.22   | 0.26   | 0.29   |        |
| Efficacy rate                        | 0.05   | 0.08   | 0.2    | 0.25   | 0.5    |        |
| Utility 2                            | -0.1   | -0.1   | 0      | 0      | 0.2    |        |
| Utility 3                            | 37     | 37.6   | 43.2   | 44.6   | 58.4   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 19.4   | 11.5   | 15     | 12.4   | 23     | 18.7   |
| U3 - Toxicity prior Beta(0.15, 0.85) | 24.3   | 12.8   | 13.9   | 9.8    | 20.6   | 18.6   |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 15.3   | 11.9   | 15.1   | 13     | 26.5   | 18.2   |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 20.5   | 12     | 14     | 11.6   | 23.5   | 18.4   |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 13.8   | 10.1   | 16     | 14.1   | 28     | 18     |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 19.7   | 11.1   | 13.5   | 12.4   | 25.1   | 18.2   |
| U2 - Toxicity prior Beta(1, 1)       | 12.4   | 9.5    | 14.3   | 14.2   | 31.7   | 17.9   |
| U3 - Toxicity prior Beta(1, 1)       | 18     | 11     | 11.6   | 13.7   | 27.5   | 18.2   |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 9.1    | 4.7    | 5.2    | 3.9    | 5      |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 8.8    | 4.8    | 5.2    | 3.9    | 5.1    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 8.6    | 4.1    | 5.2    | 4.3    | 5.7    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 8.6    | 4.2    | 5.2    | 4.4    | 5.7    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 8.3    | 3.9    | 5      | 4.7    | 6.1    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 8.3    | 3.8    | 5.1    | 4.6    | 6.1    |        |
| U2 - Toxicity prior Beta(1, 1)       | 8.2    | 3.8    | 4.7    | 4.7    | 6.6    |        |
| U3 - Toxicity prior Beta(1, 1)       | 8.2    | 3.8    | 4.7    | 4.7    | 6.5    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.6    | 2.7    | 1.3    | 1.4    | 0.7    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.5    | 2.7    | 1.3    | 1.4    | 0.7    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.4    | 2.5    | 1.3    | 1.5    | 0.7    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.4    | 2.5    | 1.4    | 1.5    | 0.7    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.4    | 2.4    | 1.3    | 1.4    | 0.7    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.3    | 2.4    | 1.3    | 1.4    | 0.7    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.3    | 2.3    | 1.1    | 1.4    | 0.6    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.3    | 2.3    | 1.1    | 1.4    | 0.6    |        |
| Scenario 10                          |        |        |        |        |        |        |
| DLT rate                             | 0.23   | 0.28   | 0.32   | 0.39   | 0.43   |        |
| Efficacy rate                        | 0.05   | 0.4    | 0.42   | 0.45   | 0.5    |        |
| Utility 2                            | -0.2   | 0.1    | 0.1    | 0.1    | 0.1    |        |
| Utility 3                            | 33.8   | 52.8   | 52.4   | 51.4   | 52.8   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 14.8   | 29.4   | 11.8   | 4.8    | 1.8    | 37.4   |
| U3 - Toxicity prior Beta(0.15, 0.85) | 14.3   | 31.7   | 10.9   | 4.4    | 1.5    | 37.2   |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 12.9   | 30.7   | 12.7   | 5.4    | 1.6    | 36.7   |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 13.4   | 31.5   | 11.6   | 5.3    | 1.5    | 36.7   |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 11.9   | 29     | 14.2   | 6.3    | 2.2    | 36.4   |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 12.9   | 30.3   | 12.9   | 5.4    | 2      | 36.5   |
| U2 - Toxicity prior Beta(1, 1)       | 12.2   | 27.8   | 14.2   | 7      | 2.8    | 36     |
| U3 - Toxicity prior Beta(1, 1)       | 12.6   | 30.3   | 12.9   | 5.7    | 2.4    | 36.1   |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 11.6   | 6.4    | 4.1    | 2.2    | 1.2    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 11.5   | 6.6    | 4      | 2.1    | 1.2    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 11.2   | 6      | 4.4    | 2.6    | 1.4    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 11.2   | 5.9    | 4.4    | 2.6    | 1.4    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 10.9   | 5.7    | 4.5    | 2.9    | 1.6    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 10.9   | 5.6    | 4.5    | 2.9    | 1.7    |        |
| U2 - Toxicity prior Beta(1, 1)       | 10.8   | 5.9    | 4.2    | 3      | 1.8    |        |
| U3 - Toxicity prior Beta(1, 1)       | 10.8   | 5.8    | 4.2    | 2.9    | 1.8    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.2    | 2.3    | 0.8    | 0.6    | 0.1    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.1    | 2.3    | 0.7    | 0.6    | 0.1    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.1    | 2.2    | 0.8    | 0.6    | 0.1    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.1    | 2.1    | 0.8    | 0.6    | 0.1    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.1    | 2.1    | 0.7    | 0.5    | 0.1    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.1    | 2.1    | 0.8    | 0.5    | 0.1    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.1    | 1.9    | 0.7    | 0.5    | 0.1    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.1    | 1.9    | 0.7    | 0.5    | 0.1    |        |

# Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two\_stage - IPDE alpha = 0.6 - both utilities - N = 30

|                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|--------------------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 11                          |        |        |        |        |        |        |
| DLT rate                             | 0.22   | 0.32   | 0.44   | 0.59   | 0.67   |        |
| Efficacy rate                        | 0.2    | 0.2    | 0.2    | 0.2    | 0.2    |        |
| Utility 2                            | 0      | -0.1   | -0.2   | -0.4   | -0.5   |        |
| Utility 3                            | 43.2   | 39.2   | 34.4   | 28.4   | 25.2   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 69.2   | 19.1   | 4.4    | 0.5    | 0      | 6.8    |
| U3 - Toxicity prior Beta(0.15, 0.85) | 71.1   | 19.3   | 2.6    | 0.4    | 0      | 6.6    |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 69.3   | 20.1   | 3.7    | 0.4    | 0      | 6.5    |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 71.7   | 18.8   | 2.6    | 0.4    | 0      | 6.5    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 68.8   | 19.2   | 4.8    | 0.5    | 0      | 6.7    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 71.5   | 18.1   | 3.2    | 0.6    | 0      | 6.6    |
| U2 - Toxicity prior Beta(1, 1)       | 69.2   | 18.8   | 4.4    | 1      | 0      | 6.6    |
| U3 - Toxicity prior Beta(1, 1)       | 72.2   | 17.4   | 3      | 0.8    | 0      | 6.6    |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 18.4   | 6.1    | 3.2    | 1      | 0.2    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 18.2   | 6.2    | 3.3    | 1      | 0.2    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 17.6   | 6.2    | 3.6    | 1.4    | 0.2    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 17.7   | 6.1    | 3.6    | 1.3    | 0.2    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 17.2   | 6.1    | 3.8    | 1.5    | 0.3    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 17.3   | 6.1    | 3.8    | 1.5    | 0.3    |        |
| U2 - Toxicity prior Beta(1, 1)       | 17.3   | 6      | 3.7    | 1.7    | 0.3    |        |
| U3 - Toxicity prior Beta(1, 1)       | 17.2   | 6      | 3.7    | 1.6    | 0.3    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.7    | 2.4    | 0.7    | 0.3    | 0      |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.7    | 2.4    | 0.7    | 0.3    | 0      |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.8    | 2.4    | 0.7    | 0.4    | 0      |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.8    | 2.3    | 0.7    | 0.4    | 0      |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.7    | 2.2    | 0.6    | 0.3    | 0      |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.7    | 2.2    | 0.6    | 0.3    | 0      |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.6    | 1.9    | 0.5    | 0.3    | 0      |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.6    | 1.9    | 0.5    | 0.3    | 0      |        |
| Scenario 12                          |        |        |        |        |        |        |
| DLT rate                             | 0.06   | 0.09   | 0.18   | 0.29   | 0.44   |        |
| Efficacy rate                        | 0.05   | 0.1    | 0.12   | 0.4    | 0.6    |        |
| Utility 2                            | 0      | 0      | -0.1   | 0.1    | 0.2    |        |
| Utility 3                            | 40.6   | 42.4   | 40     | 52.4   | 58.4   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 16.6   | 14.8   | 11.3   | 34.7   | 19.4   | 3.2    |
| U3 - Toxicity prior Beta(0.15, 0.85) | 24.5   | 15.9   | 7.6    | 31.3   | 17.4   | 3.3    |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 16.3   | 12.7   | 9.8    | 36.3   | 21.7   | 3.2    |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 22.3   | 13.1   | 8.2    | 33.9   | 19.3   | 3.2    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 14.2   | 12.1   | 9.1    | 39.1   | 22.3   | 3.2    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 19.3   | 14.1   | 6.5    | 36.8   | 20     | 3.3    |
| U2 - Toxicity prior Beta(1, 1)       | 14.8   | 12.1   | 9.1    | 36     | 24.8   | 3.2    |
| U3 - Toxicity prior Beta(1, 1)       | 18.9   | 13.5   | 7      | 33.8   | 23.5   | 3.3    |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 5      | 5      | 5.3    | 7.4    | 6.9    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 5.1    | 4.9    | 5.4    | 7.3    | 6.9    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 4.9    | 4.4    | 5.2    | 7.9    | 7.3    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 4.9    | 4.4    | 5.2    | 7.7    | 7.3    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 4.7    | 4.3    | 4.9    | 8      | 7.6    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 4.7    | 4.3    | 4.9    | 8      | 7.6    |        |
| U2 - Toxicity prior Beta(1, 1)       | 4.7    | 4.2    | 4.8    | 7.7    | 8.3    |        |
| U3 - Toxicity prior Beta(1, 1)       | 4.7    | 4.2    | 4.8    | 7.6    | 8.4    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.4    | 3      | 1.4    | 2.2    | 0.8    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.4    | 3      | 1.4    | 2.3    | 0.8    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.3    | 2.8    | 1.4    | 2.3    | 0.8    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.3    | 2.9    | 1.4    | 2.3    | 0.8    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.3    | 2.8    | 1.3    | 2.2    | 0.8    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.3    | 2.8    | 1.3    | 2.2    | 0.8    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.3    | 2.7    | 1.2    | 2.1    | 0.8    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.3    | 2.7    | 1.2    | 2.1    | 0.7    |        |

# Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two\_stage - IPDE alpha = 0.6 - both utilities - N = 30

|                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|--------------------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 13                          |        |        |        |        |        |        |
| DLT rate                             | 0.21   | 0.29   | 0.3    | 0.39   | 0.46   |        |
| Efficacy rate                        | 0.05   | 0.3    | 0.32   | 0.35   | 0.4    |        |
| Utility 2                            | -0.2   | 0      | 0      | 0      | -0.1   |        |
| Utility 3                            | 34.6   | 46.4   | 47.2   | 45.4   | 45.6   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 16.5   | 29.8   | 14.1   | 4.9    | 2      | 32.7   |
| U3 - Toxicity prior Beta(0.15, 0.85) | 17.5   | 32.4   | 11     | 5.1    | 1      | 33     |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 15.4   | 30.7   | 13.3   | 5.5    | 2.7    | 32.4   |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 15.8   | 32     | 12.8   | 5.2    | 1.9    | 32.3   |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 13.9   | 31.2   | 14.4   | 5.7    | 3.3    | 31.5   |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 15.6   | 31.7   | 12.7   | 5.6    | 2.7    | 31.7   |
| U2 - Toxicity prior Beta(1, 1)       | 14.2   | 29.7   | 14.9   | 6.7    | 3.1    | 31.4   |
| U3 - Toxicity prior Beta(1, 1)       | 15.4   | 31.9   | 13.3   | 5.4    | 2.6    | 31.4   |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 11.2   | 6.5    | 4.5    | 2.4    | 1.4    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 11.2   | 6.5    | 4.4    | 2.5    | 1.4    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 10.8   | 6.3    | 4.6    | 2.9    | 1.6    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 10.7   | 6.2    | 4.6    | 2.9    | 1.6    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 10.4   | 6      | 4.7    | 3.3    | 1.8    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 10.5   | 6      | 4.6    | 3.3    | 1.8    |        |
| U2 - Toxicity prior Beta(1, 1)       | 10.4   | 6.1    | 4.5    | 3.3    | 2      |        |
| U3 - Toxicity prior Beta(1, 1)       | 10.4   | 6      | 4.5    | 3.3    | 2      |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.3    | 2.6    | 0.9    | 0.7    | 0.2    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.2    | 2.6    | 0.9    | 0.7    | 0.2    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.2    | 2.4    | 0.9    | 0.7    | 0.2    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.2    | 2.4    | 0.9    | 0.7    | 0.2    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.1    | 2.3    | 0.8    | 0.7    | 0.2    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.2    | 2.3    | 0.8    | 0.7    | 0.2    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.1    | 2.1    | 0.7    | 0.7    | 0.2    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.1    | 2.1    | 0.7    | 0.7    | 0.2    |        |
| Scenario 14                          |        |        |        |        |        |        |
| DLT rate                             | 0.09   | 0.15   | 0.23   | 0.28   | 0.39   |        |
| Efficacy rate                        | 0.05   | 0.08   | 0.12   | 0.3    | 0.4    |        |
| Utility 2                            | 0      | -0.1   | -0.1   | 0      | 0      |        |
| Utility 3                            | 39.4   | 38.8   | 38     | 46.8   | 48.4   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 24.3   | 16.5   | 15.2   | 22.8   | 13.7   | 7.5    |
| U3 - Toxicity prior Beta(0.15, 0.85) | 32     | 16.4   | 11.5   | 21.5   | 11.1   | 7.5    |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 20.4   | 15.6   | 14.2   | 26     | 16.6   | 7.2    |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 29.6   | 16.9   | 10.5   | 23.2   | 12.6   | 7.2    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 19.5   | 14     | 13.7   | 27.5   | 18.1   | 7.2    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 29.2   | 16.4   | 9.7    | 24.1   | 13.4   | 7.2    |
| U2 - Toxicity prior Beta(1, 1)       | 19.5   | 12.7   | 13.3   | 28     | 19.3   | 7.2    |
| U3 - Toxicity prior Beta(1, 1)       | 27.7   | 15     | 10.1   | 25.1   | 14.8   | 7.3    |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 7      | 5.6    | 5.8    | 5.8    | 5      |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 7      | 5.7    | 5.6    | 5.8    | 5.1    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 6.6    | 4.9    | 5.5    | 6.5    | 5.6    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 6.6    | 5      | 5.4    | 6.5    | 5.6    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 6.4    | 4.7    | 5.2    | 6.9    | 5.9    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 6.4    | 4.7    | 5.2    | 6.8    | 5.9    |        |
| U2 - Toxicity prior Beta(1, 1)       | 6.3    | 4.6    | 5.1    | 6.7    | 6.5    |        |
| U3 - Toxicity prior Beta(1, 1)       | 6.3    | 4.7    | 5      | 6.7    | 6.5    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.7    | 3.2    | 1.5    | 1.9    | 0.7    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.6    | 3.2    | 1.5    | 2      | 0.7    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.5    | 3      | 1.5    | 2.1    | 0.8    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.6    | 2.9    | 1.4    | 2      | 0.8    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.5    | 2.9    | 1.4    | 2      | 0.7    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.5    | 2.9    | 1.4    | 1.9    | 0.7    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.5    | 2.8    | 1.3    | 1.9    | 0.7    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.5    | 2.8    | 1.3    | 1.9    | 0.7    |        |

# Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two\_stage - IPDE alpha = 0.6 - both utilities - N = 30

|                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|--------------------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 15                          |        |        |        |        |        |        |
| DLT rate                             | 0.13   | 0.2    | 0.32   | 0.39   | 0.46   |        |
| Efficacy rate                        | 0.1    | 0.1    | 0.1    | 0.1    | 0.1    |        |
| Utility 2                            | 0      | -0.1   | -0.2   | -0.3   | -0.4   |        |
| Utility 3                            | 40.8   | 38     | 33.2   | 30.4   | 27.6   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 51.8   | 28.5   | 8.8    | 2.7    | 1.1    | 7.1    |
| U3 - Toxicity prior Beta(0.15, 0.85) | 60.9   | 23.6   | 6.4    | 1.1    | 0.7    | 7.3    |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 50.6   | 28.2   | 9.8    | 3.4    | 0.9    | 7.1    |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 62.1   | 23.3   | 5.2    | 1.5    | 0.8    | 7.1    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 51.1   | 28     | 9.8    | 2.9    | 1.3    | 6.9    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 62.9   | 23.6   | 4.7    | 1.3    | 0.6    | 6.9    |
| U2 - Toxicity prior Beta(1, 1)       | 51.2   | 26.6   | 9.8    | 4.5    | 0.9    | 7      |
| U3 - Toxicity prior Beta(1, 1)       | 62.9   | 22.5   | 5.2    | 1.8    | 0.6    | 7      |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 11.3   | 7.4    | 5.6    | 3.2    | 1.7    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 11.3   | 7.4    | 5.6    | 3.2    | 1.7    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 10.9   | 6.9    | 5.7    | 3.8    | 1.9    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 11     | 7      | 5.7    | 3.7    | 1.9    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 10.8   | 6.7    | 5.6    | 4      | 2      |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 10.9   | 6.6    | 5.6    | 4      | 2.1    |        |
| U2 - Toxicity prior Beta(1, 1)       | 10.7   | 6.5    | 5.5    | 4.2    | 2.4    |        |
| U3 - Toxicity prior Beta(1, 1)       | 10.7   | 6.6    | 5.4    | 4.1    | 2.4    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 1.1    | 3.5    | 1.5    | 1.3    | 0.4    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 1.1    | 3.5    | 1.5    | 1.3    | 0.4    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 1.1    | 3.2    | 1.5    | 1.4    | 0.4    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 1.1    | 3.2    | 1.5    | 1.4    | 0.4    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 1.1    | 3.2    | 1.4    | 1.3    | 0.4    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 1.1    | 3.2    | 1.4    | 1.3    | 0.4    |        |
| U2 - Toxicity prior Beta(1, 1)       | 1.1    | 3      | 1.2    | 1.2    | 0.4    |        |
| U3 - Toxicity prior Beta(1, 1)       | 1.1    | 3.1    | 1.2    | 1.3    | 0.3    |        |
| Scenario 16                          |        |        |        |        |        |        |
| DLT rate                             | 0.02   | 0.06   | 0.09   | 0.14   | 0.26   |        |
| Efficacy rate                        | 0.2    | 0.2    | 0.2    | 0.2    | 0.2    |        |
| Utility 2                            | 0.2    | 0.1    | 0.1    | 0.1    | -0.1   |        |
| Utility 3                            | 51.2   | 49.6   | 48.4   | 46.4   | 41.6   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 28.6   | 26.3   | 17.8   | 16.7   | 10.5   | 0.1    |
| U3 - Toxicity prior Beta(0.15, 0.85) | 35.5   | 29.7   | 16.1   | 13.1   | 5.5    | 0.1    |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 28.5   | 25.6   | 17     | 18     | 10.8   | 0.1    |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 35.7   | 30.4   | 15.3   | 13.3   | 5.2    | 0.1    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 28.3   | 27.8   | 16     | 17     | 10.8   | 0.1    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 35.5   | 31.1   | 14.8   | 12.7   | 5.8    | 0.1    |
| U2 - Toxicity prior Beta(1, 1)       | 26.8   | 25.9   | 17.6   | 17.7   | 11.9   | 0.1    |
| U3 - Toxicity prior Beta(1, 1)       | 34.9   | 30.1   | 15.6   | 13.2   | 6.1    | 0.1    |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 5.9    | 5.7    | 5.5    | 5.8    | 7.2    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 6      | 5.7    | 5.3    | 5.7    | 7.2    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 5.9    | 5.6    | 5.3    | 5.8    | 7.3    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 5.9    | 5.6    | 5.3    | 5.8    | 7.3    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 5.9    | 5.6    | 5.3    | 5.7    | 7.4    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 5.9    | 5.6    | 5.3    | 5.8    | 7.4    |        |
| U2 - Toxicity prior Beta(1, 1)       | 5.9    | 5.6    | 5.2    | 5.8    | 7.5    |        |
| U3 - Toxicity prior Beta(1, 1)       | 5.9    | 5.5    | 5.2    | 5.8    | 7.5    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 1.1    | 3.2    | 1.8    | 2.6    | 1.5    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 1.1    | 3.2    | 1.8    | 2.6    | 1.4    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 1.1    | 3.2    | 1.8    | 2.6    | 1.5    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 1.1    | 3.2    | 1.8    | 2.7    | 1.5    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 1.1    | 3.2    | 1.7    | 2.6    | 1.5    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 1.1    | 3.2    | 1.8    | 2.6    | 1.5    |        |
| U2 - Toxicity prior Beta(1, 1)       | 1.1    | 3.2    | 1.8    | 2.6    | 1.4    |        |
| U3 - Toxicity prior Beta(1, 1)       | 1.1    | 3.2    | 1.8    | 2.6    | 1.5    |        |



# Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two\_stage - IPDE alpha = 0.6 - both utilities - N = 30

|                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|--------------------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 17                          |        |        |        |        |        |        |
| DLT rate                             | 0.11   | 0.23   | 0.39   | 0.58   | 0.76   |        |
| Efficacy rate                        | 0.3    | 0.5    | 0.5    | 0.5    | 0.5    |        |
| Utility 2                            | 0.2    | 0.3    | 0.1    | -0.1   | -0.3   |        |
| Utility 3                            | 53.6   | 60.8   | 54.4   | 46.8   | 39.6   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 32     | 50.3   | 16     | 1.4    | 0      | 0.3    |
| U3 - Toxicity prior Beta(0.15, 0.85) | 35.3   | 49     | 14     | 1.4    | 0      | 0.3    |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 31.2   | 50.4   | 16.2   | 1.9    | 0      | 0.3    |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 34     | 50.6   | 13.9   | 1.2    | 0      | 0.3    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 30.7   | 50.1   | 16.4   | 2.5    | 0      | 0.3    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 32.5   | 50.4   | 14.7   | 2.1    | 0      | 0.3    |
| U2 - Toxicity prior Beta(1, 1)       | 30.2   | 50.3   | 17.1   | 2.1    | 0      | 0.3    |
| U3 - Toxicity prior Beta(1, 1)       | 33     | 50     | 14.5   | 2.2    | 0      | 0.3    |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 10.1   | 10.2   | 6.9    | 2.2    | 0.5    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 10     | 10.2   | 7.1    | 2.3    | 0.4    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 9.5    | 9.9    | 7.2    | 2.8    | 0.5    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 9.5    | 10     | 7.2    | 2.8    | 0.5    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 9.2    | 10.1   | 7.1    | 3.1    | 0.6    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 9.3    | 9.9    | 7.1    | 3.1    | 0.6    |        |
| U2 - Toxicity prior Beta(1, 1)       | 9.2    | 10     | 7      | 3.2    | 0.6    |        |
| U3 - Toxicity prior Beta(1, 1)       | 9.1    | 10.1   | 7      | 3.2    | 0.6    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.4    | 2.9    | 1.1    | 0.6    | 0.1    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.4    | 2.8    | 1.1    | 0.6    | 0.1    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.5    | 2.7    | 1.1    | 0.6    | 0.1    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.4    | 2.8    | 1.1    | 0.6    | 0.1    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.4    | 2.6    | 1      | 0.6    | 0.1    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.4    | 2.6    | 1      | 0.6    | 0.1    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.4    | 2.5    | 0.9    | 0.5    | 0.1    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.4    | 2.5    | 0.9    | 0.5    | 0.1    |        |
| Scenario 18                          |        |        |        |        |        |        |
| DLT rate                             | 0.02   | 0.04   | 0.1    | 0.2    | 0.35   |        |
| Efficacy rate                        | 0.05   | 0.1    | 0.15   | 0.2    | 0.5    |        |
| Utility 2                            | 0      | 0.1    | 0.1    | 0      | 0.1    |        |
| Utility 3                            | 42.2   | 44.4   | 45     | 44     | 56     |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 10.1   | 8.9    | 12.2   | 17.9   | 50.2   | 0.7    |
| U3 - Toxicity prior Beta(0.15, 0.85) | 15.6   | 14.1   | 10.4   | 15.4   | 43.8   | 0.7    |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 9.8    | 9.3    | 11.7   | 16.9   | 51.6   | 0.7    |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 15.4   | 13.5   | 11     | 14.5   | 44.9   | 0.7    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 10.3   | 8.7    | 11.4   | 15.7   | 53.2   | 0.7    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 15.1   | 12.1   | 10.7   | 14     | 47.4   | 0.7    |
| U2 - Toxicity prior Beta(1, 1)       | 9.3    | 8.5    | 10.5   | 14.8   | 56.2   | 0.7    |
| U3 - Toxicity prior Beta(1, 1)       | 14.2   | 12.6   | 9.5    | 13     | 50     | 0.7    |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 3.8    | 3.9    | 4.7    | 5.7    | 11.8   |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 3.8    | 3.8    | 4.8    | 5.7    | 11.8   |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 3.8    | 3.8    | 4.5    | 5.7    | 12.1   |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 3.8    | 3.8    | 4.6    | 5.7    | 12     |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 3.8    | 3.8    | 4.5    | 5.6    | 12.3   |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 3.8    | 3.8    | 4.5    | 5.7    | 12.2   |        |
| U2 - Toxicity prior Beta(1, 1)       | 3.7    | 3.7    | 4.4    | 5.5    | 12.5   |        |
| U3 - Toxicity prior Beta(1, 1)       | 3.7    | 3.7    | 4.4    | 5.5    | 12.5   |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.2    | 2.8    | 1.4    | 2.4    | 1.4    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.3    | 2.7    | 1.4    | 2.3    | 1.4    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.2    | 2.7    | 1.4    | 2.4    | 1.5    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.2    | 2.7    | 1.4    | 2.4    | 1.4    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.2    | 2.7    | 1.4    | 2.3    | 1.4    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.2    | 2.7    | 1.4    | 2.3    | 1.4    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.2    | 2.7    | 1.4    | 2.3    | 1.4    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.2    | 2.7    | 1.3    | 2.3    | 1.4    |        |

# Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two\_stage - IPDE alpha = 0.6 - both utilities - N = 30

|                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|--------------------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 19                          |        |        |        |        |        |        |
| DLT rate                             | 0.11   | 0.26   | 0.39   | 0.6    | 0.77   |        |
| Efficacy rate                        | 0.08   | 0.3    | 0.32   | 0.35   | 0.4    |        |
| Utility 2                            | 0      | 0      | -0.1   | -0.2   | -0.4   |        |
| Utility 3                            | 40.4   | 47.6   | 43.6   | 37     | 33.2   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 24.3   | 50.9   | 14.6   | 1.7    | 0      | 8.5    |
| U3 - Toxicity prior Beta(0.15, 0.85) | 26.9   | 50.1   | 13.2   | 1.2    | 0      | 8.6    |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 25.9   | 47.4   | 16     | 2.2    | 0      | 8.5    |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 26.5   | 49.6   | 13.6   | 1.6    | 0.1    | 8.6    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 24.8   | 48.4   | 16.7   | 1.8    | 0.1    | 8.2    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 27.9   | 47.4   | 14.6   | 1.6    | 0.1    | 8.4    |
| U2 - Toxicity prior Beta(1, 1)       | 24.4   | 48.1   | 16.6   | 2.6    | 0.1    | 8.2    |
| U3 - Toxicity prior Beta(1, 1)       | 25.8   | 48.7   | 15.3   | 1.9    | 0.1    | 8.2    |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 9.2    | 10.4   | 6.8    | 2.2    | 0.5    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 9.1    | 10.5   | 6.8    | 2.2    | 0.5    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 8.8    | 9.9    | 7.2    | 2.8    | 0.5    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 8.7    | 9.9    | 7.1    | 2.9    | 0.5    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 8.6    | 9.6    | 7.2    | 3.2    | 0.6    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 8.7    | 9.6    | 7.2    | 3.2    | 0.6    |        |
| U2 - Toxicity prior Beta(1, 1)       | 8.5    | 9.5    | 7.2    | 3.4    | 0.7    |        |
| U3 - Toxicity prior Beta(1, 1)       | 8.4    | 9.5    | 7.2    | 3.4    | 0.7    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.5    | 3.6    | 1.2    | 0.7    | 0.1    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.5    | 3.6    | 1.2    | 0.7    | 0.1    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.5    | 3.3    | 1.2    | 0.8    | 0.1    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.5    | 3.3    | 1.2    | 0.8    | 0.1    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.5    | 3.2    | 1.1    | 0.7    | 0.1    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.5    | 3.2    | 1.1    | 0.7    | 0.1    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.4    | 3      | 1      | 0.7    | 0.1    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.4    | 2.9    | 1      | 0.7    | 0.1    |        |
| Scenario 20                          |        |        |        |        |        |        |
| DLT rate                             | 0.07   | 0.14   | 0.31   | 0.41   | 0.52   |        |
| Efficacy rate                        | 0.05   | 0.25   | 0.25   | 0.25   | 0.25   |        |
| Utility 2                            | 0      | 0.1    | -0.1   | -0.2   | -0.3   |        |
| Utility 3                            | 40.2   | 49.4   | 42.6   | 38.6   | 34.2   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 14.6   | 45.8   | 27.8   | 6.5    | 1      | 4.3    |
| U3 - Toxicity prior Beta(0.15, 0.85) | 20.5   | 47     | 22.3   | 5.6    | 0.3    | 4.3    |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 14.6   | 43.2   | 27     | 9.7    | 1.2    | 4.3    |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 19.7   | 46.2   | 22     | 7      | 0.8    | 4.3    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 15     | 42.7   | 26.1   | 9.9    | 2.1    | 4.2    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 19.5   | 45.2   | 22.4   | 8.1    | 0.6    | 4.2    |
| U2 - Toxicity prior Beta(1, 1)       | 14.7   | 38.9   | 27.7   | 12.2   | 2.3    | 4.2    |
| U3 - Toxicity prior Beta(1, 1)       | 18.6   | 44.1   | 23.1   | 9      | 1      | 4.2    |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 5.8    | 8.7    | 8.1    | 4.6    | 2.1    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 5.9    | 8.7    | 8.2    | 4.6    | 2.1    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 5.7    | 8.1    | 7.8    | 5.4    | 2.4    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 5.6    | 8.1    | 7.9    | 5.4    | 2.4    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 5.5    | 7.8    | 7.7    | 5.8    | 2.6    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 5.5    | 7.7    | 7.9    | 5.8    | 2.6    |        |
| U2 - Toxicity prior Beta(1, 1)       | 5.4    | 7.6    | 7.6    | 5.9    | 3      |        |
| U3 - Toxicity prior Beta(1, 1)       | 5.4    | 7.5    | 7.6    | 5.9    | 3      |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.4    | 3.6    | 1.7    | 1.6    | 0.5    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.5    | 3.6    | 1.8    | 1.5    | 0.4    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.4    | 3.5    | 1.7    | 1.6    | 0.5    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.4    | 3.4    | 1.8    | 1.6    | 0.5    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.4    | 3.4    | 1.7    | 1.6    | 0.5    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.4    | 3.4    | 1.8    | 1.6    | 0.5    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.4    | 3.4    | 1.5    | 1.5    | 0.4    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.4    | 3.3    | 1.6    | 1.5    | 0.4    |        |

# Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two\_stage - IPDE alpha = 0.6 - both utilities - N = 30

|                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|--------------------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 21                          |        |        |        |        |        |        |
| DLT rate                             | 0.37   | 0.56   | 0.69   | 0.83   | 0.88   |        |
| Efficacy rate                        | 0.5    | 0.5    | 0.5    | 0.5    | 0.5    |        |
| Utility 2                            | 0.1    | -0.1   | -0.2   | -0.3   | -0.4   |        |
| Utility 3                            | 55.2   | 47.6   | 42.4   | 36.8   | 34.8   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 58     | 1.6    | 0      | 0      | 0      | 40.4   |
| U3 - Toxicity prior Beta(0.15, 0.85) | 58.1   | 1.5    | 0      | 0      | 0      | 40.4   |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 59.4   | 1.5    | 0      | 0      | 0      | 39.1   |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 58.6   | 1.5    | 0      | 0      | 0      | 39.9   |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 59.5   | 1.4    | 0.1    | 0      | 0      | 39     |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 59.7   | 1.3    | 0      | 0      | 0      | 39     |
| U2 - Toxicity prior Beta(1, 1)       | 59.7   | 1.5    | 0.1    | 0      | 0      | 38.7   |
| U3 - Toxicity prior Beta(1, 1)       | 59.7   | 1.3    | 0      | 0      | 0      | 39     |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 20.5   | 2.2    | 0.6    | 0.1    | 0      |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 20.6   | 2.2    | 0.6    | 0.1    | 0      |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 20.4   | 2.5    | 0.6    | 0.1    | 0      |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 20.3   | 2.5    | 0.6    | 0.1    | 0      |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 20.3   | 2.5    | 0.7    | 0.1    | 0      |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 20.3   | 2.6    | 0.6    | 0.1    | 0      |        |
| U2 - Toxicity prior Beta(1, 1)       | 20.3   | 2.6    | 0.7    | 0.1    | 0      |        |
| U3 - Toxicity prior Beta(1, 1)       | 20.2   | 2.6    | 0.7    | 0.1    | 0      |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.2    | 0.7    | 0.1    | 0      | 0      |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.1    | 0.7    | 0.1    | 0      | 0      |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.1    | 0.6    | 0.1    | 0      | 0      |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.1    | 0.6    | 0.1    | 0      | 0      |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.1    | 0.6    | 0.1    | 0      | 0      |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.1    | 0.6    | 0.1    | 0      | 0      |        |
| U2 - Toxicity prior Beta(1, 1)       | 0      | 0.6    | 0.1    | 0      | 0      |        |
| U3 - Toxicity prior Beta(1, 1)       | 0      | 0.6    | 0.1    | 0      | 0      |        |
| Scenario 22                          |        |        |        |        |        |        |
| DLT rate                             | 0.07   | 0.27   | 0.5    | 0.56   | 0.78   |        |
| Efficacy rate                        | 0.08   | 0.25   | 0.25   | 0.5    | 0.5    |        |
| Utility 2                            | 0      | 0      | -0.2   | -0.1   | -0.3   |        |
| Utility 3                            | 42     | 44.2   | 35     | 47.6   | 38.8   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 28.6   | 53.1   | 12.6   | 1      | 0      | 4.7    |
| U3 - Toxicity prior Beta(0.15, 0.85) | 31.8   | 52.5   | 9.9    | 1.1    | 0      | 4.7    |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 29.8   | 49.9   | 14.3   | 1.6    | 0      | 4.4    |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 32.8   | 50.6   | 10.4   | 1.6    | 0      | 4.6    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 32.4   | 48.3   | 12.8   | 2.3    | 0      | 4.2    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 35.3   | 48.8   | 9.7    | 2      | 0      | 4.2    |
| U2 - Toxicity prior Beta(1, 1)       | 29.8   | 49.7   | 13.5   | 2.8    | 0.1    | 4.1    |
| U3 - Toxicity prior Beta(1, 1)       | 34.8   | 48.4   | 10.1   | 2.6    | 0      | 4.1    |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 9.3    | 11.9   | 6      | 1.8    | 0.3    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 9.3    | 12     | 6.1    | 1.7    | 0.3    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 8.8    | 11.5   | 6.3    | 2.5    | 0.4    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 9      | 11.2   | 6.4    | 2.5    | 0.4    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 8.5    | 11     | 6.6    | 2.9    | 0.5    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 8.5    | 11.1   | 6.6    | 2.9    | 0.5    |        |
| U2 - Toxicity prior Beta(1, 1)       | 8.3    | 11     | 6.6    | 3.1    | 0.6    |        |
| U3 - Toxicity prior Beta(1, 1)       | 8.3    | 10.9   | 6.7    | 3      | 0.6    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.8    | 3.9    | 1.2    | 0.5    | 0      |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.8    | 3.9    | 1.2    | 0.5    | 0      |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.8    | 3.6    | 1.1    | 0.6    | 0      |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.8    | 3.6    | 1.1    | 0.6    | 0      |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.7    | 3.4    | 1      | 0.6    | 0      |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.7    | 3.4    | 1.1    | 0.6    | 0      |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.7    | 3.2    | 0.9    | 0.6    | 0      |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.7    | 3.3    | 0.9    | 0.5    | 0      |        |

# Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two\_stage - IPDE alpha = 0.6 - both utilities - N = 30

|                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|--------------------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 23                          |        |        |        |        |        |        |
| DLT rate                             | 0.32   | 0.55   | 0.8    | 0.94   | 0.97   |        |
| Efficacy rate                        | 0.05   | 0.05   | 0.05   | 0.05   | 0.05   |        |
| Utility 2                            | -0.3   | -0.5   | -0.8   | -0.9   | -0.9   |        |
| Utility 3                            | 30.2   | 21     | 11     | 5.4    | 4.2    |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 21     | 3.7    | 0.2    | 0      | 0      | 75.1   |
| U3 - Toxicity prior Beta(0.15, 0.85) | 21.3   | 3.3    | 0.2    | 0      | 0      | 75.2   |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 21.1   | 3.1    | 0.5    | 0      | 0      | 75.3   |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 21.1   | 3.3    | 0.4    | 0      | 0      | 75.2   |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 21.2   | 3.9    | 0.4    | 0      | 0      | 74.5   |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 21.3   | 3.9    | 0.2    | 0      | 0      | 74.6   |
| U2 - Toxicity prior Beta(1, 1)       | 20.8   | 3.5    | 0.5    | 0      | 0      | 75.2   |
| U3 - Toxicity prior Beta(1, 1)       | 21     | 3.7    | 0      | 0      | 0      | 75.3   |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 16.6   | 3.1    | 0.9    | 0.1    | 0      |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 16.7   | 3.1    | 0.9    | 0.1    | 0      |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 16.6   | 3.2    | 1      | 0.1    | 0      |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 16.6   | 3.2    | 1      | 0.1    | 0      |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 16.7   | 3.1    | 1.1    | 0.1    | 0      |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 16.7   | 3.2    | 1.1    | 0.1    | 0      |        |
| U2 - Toxicity prior Beta(1, 1)       | 16.6   | 3.2    | 1.1    | 0.1    | 0      |        |
| U3 - Toxicity prior Beta(1, 1)       | 16.7   | 3.2    | 1.1    | 0.1    | 0      |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.4    | 1.4    | 0.2    | 0      | 0      |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.4    | 1.4    | 0.2    | 0      | 0      |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.3    | 1.3    | 0.1    | 0      | 0      |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.3    | 1.3    | 0.1    | 0      | 0      |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.2    | 1.2    | 0.1    | 0      | 0      |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.2    | 1.2    | 0.1    | 0      | 0      |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.1    | 1.1    | 0.1    | 0      | 0      |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.1    | 1.1    | 0.1    | 0      | 0      |        |
| Scenario 24                          |        |        |        |        |        |        |
| DLT rate                             | 0.01   | 0.02   | 0.05   | 0.1    | 0.25   |        |
| Efficacy rate                        | 0.05   | 0.08   | 0.2    | 0.5    | 0.5    |        |
| Utility 2                            | 0      | 0.1    | 0.1    | 0.4    | 0.2    |        |
| Utility 3                            | 42.6   | 44     | 50     | 66     | 60     |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 1.3    | 1.1    | 6.3    | 46.9   | 44.4   | 0      |
| U3 - Toxicity prior Beta(0.15, 0.85) | 2.4    | 2.4    | 7.6    | 47.5   | 40.1   | 0      |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 1      | 1.8    | 6.4    | 45.8   | 45     | 0      |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 2.4    | 2.5    | 7.8    | 47.2   | 40.1   | 0      |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 1      | 1.2    | 6.5    | 45.5   | 45.8   | 0      |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 2.2    | 2.2    | 7.8    | 47.1   | 40.7   | 0      |
| U2 - Toxicity prior Beta(1, 1)       | 0.9    | 1.7    | 6      | 45.1   | 46.3   | 0      |
| U3 - Toxicity prior Beta(1, 1)       | 2.7    | 2.3    | 7.4    | 47     | 40.6   | 0      |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 3.3    | 3.2    | 3.8    | 8.5    | 11.3   |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 3.3    | 3.2    | 3.7    | 8.4    | 11.4   |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 3.3    | 3.1    | 3.7    | 8.4    | 11.5   |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 3.3    | 3.1    | 3.7    | 8.4    | 11.6   |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 3.3    | 3.2    | 3.7    | 8.4    | 11.5   |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 3.3    | 3.2    | 3.7    | 8.4    | 11.6   |        |
| U2 - Toxicity prior Beta(1, 1)       | 3.3    | 3.1    | 3.7    | 8.3    | 11.6   |        |
| U3 - Toxicity prior Beta(1, 1)       | 3.3    | 3.1    | 3.7    | 8.4    | 11.5   |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.1    | 2.5    | 1.2    | 3.2    | 1.2    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.1    | 2.5    | 1.2    | 3.2    | 1.2    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.1    | 2.5    | 1.2    | 3.2    | 1.2    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.1    | 2.5    | 1.2    | 3.2    | 1.2    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.1    | 2.5    | 1.2    | 3.1    | 1.2    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.1    | 2.5    | 1.1    | 3.1    | 1.2    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.1    | 2.5    | 1.2    | 3.1    | 1.2    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.1    | 2.5    | 1.2    | 3.1    | 1.2    |        |



# Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two\_stage - IPDE alpha = 0.6 - both utilities - N = 30

|                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|--------------------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 25                          |        |        |        |        |        |        |
| DLT rate                             | 0.09   | 0.18   | 0.37   | 0.63   | 0.77   |        |
| Efficacy rate                        | 0.2    | 0.25   | 0.4    | 0.45   | 0.5    |        |
| Utility 2                            | 0.1    | 0.1    | 0      | -0.2   | -0.3   |        |
| Utility 3                            | 48.4   | 47.8   | 49.2   | 41.8   | 39.2   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 35.9   | 31.6   | 28     | 3.8    | 0      | 0.7    |
| U3 - Toxicity prior Beta(0.15, 0.85) | 41.9   | 30.5   | 23.9   | 3      | 0      | 0.7    |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 36     | 30.4   | 28.3   | 4.5    | 0.1    | 0.7    |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 41.5   | 29.5   | 24.9   | 3.3    | 0.1    | 0.7    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 34.5   | 29.5   | 29.4   | 5.8    | 0.1    | 0.7    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 40.6   | 29.6   | 24.8   | 4.3    | 0      | 0.7    |
| U2 - Toxicity prior Beta(1, 1)       | 34.6   | 31.4   | 28.2   | 5      | 0.1    | 0.7    |
| U3 - Toxicity prior Beta(1, 1)       | 40.5   | 30.7   | 23.9   | 4.1    | 0.1    | 0.7    |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 9.6    | 8.3    | 8.6    | 2.8    | 0.5    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 9.5    | 8.4    | 8.7    | 2.8    | 0.5    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 9.2    | 7.8    | 8.8    | 3.5    | 0.6    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 9.2    | 7.9    | 8.8    | 3.5    | 0.6    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 9      | 7.8    | 8.6    | 3.9    | 0.6    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 9      | 7.8    | 8.5    | 3.8    | 0.7    |        |
| U2 - Toxicity prior Beta(1, 1)       | 8.8    | 7.9    | 8.5    | 4      | 0.8    |        |
| U3 - Toxicity prior Beta(1, 1)       | 8.7    | 7.8    | 8.6    | 4      | 0.8    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.8    | 3.1    | 1.5    | 0.8    | 0.1    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.8    | 3.2    | 1.5    | 0.8    | 0.1    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.8    | 3      | 1.5    | 0.9    | 0.1    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.8    | 3      | 1.5    | 0.9    | 0.1    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.7    | 2.9    | 1.3    | 0.8    | 0.1    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.8    | 2.9    | 1.3    | 0.8    | 0.1    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.7    | 2.8    | 1.2    | 0.8    | 0.1    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.7    | 2.8    | 1.2    | 0.8    | 0.1    |        |
| Scenario 26                          |        |        |        |        |        |        |
| DLT rate                             | 0.15   | 0.22   | 0.3    | 0.62   | 0.8    |        |
| Efficacy rate                        | 0.1    | 0.2    | 0.35   | 0.45   | 0.55   |        |
| Utility 2                            | -0.1   | 0      | 0.1    | -0.2   | -0.2   |        |
| Utility 3                            | 40     | 43.2   | 49     | 42.2   | 41     |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 34.2   | 27.2   | 23.4   | 4.6    | 0      | 10.6   |
| U3 - Toxicity prior Beta(0.15, 0.85) | 36.9   | 28.3   | 19.6   | 4.7    | 0      | 10.5   |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 35.1   | 25.6   | 23.9   | 5.4    | 0      | 10     |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 36.9   | 26.3   | 21.9   | 5      | 0      | 9.9    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 33.5   | 25.8   | 25.4   | 5.6    | 0      | 9.7    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 37.1   | 25.2   | 22.3   | 5.6    | 0      | 9.8    |
| U2 - Toxicity prior Beta(1, 1)       | 33.1   | 26.3   | 23.7   | 7      | 0.1    | 9.8    |
| U3 - Toxicity prior Beta(1, 1)       | 36.5   | 26     | 21.3   | 6.4    | 0      | 9.8    |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 11.1   | 7.3    | 7.2    | 2.8    | 0.5    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 10.9   | 7.4    | 7.4    | 2.8    | 0.5    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 10.6   | 7      | 7.5    | 3.4    | 0.6    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 10.5   | 6.9    | 7.5    | 3.5    | 0.6    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 10.4   | 6.6    | 7.5    | 3.8    | 0.7    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 10.4   | 6.6    | 7.5    | 3.8    | 0.7    |        |
| U2 - Toxicity prior Beta(1, 1)       | 10.3   | 6.6    | 7.5    | 3.9    | 0.8    |        |
| U3 - Toxicity prior Beta(1, 1)       | 10.3   | 6.6    | 7.5    | 3.9    | 0.8    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.6    | 3      | 1.2    | 0.8    | 0.1    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.5    | 3      | 1.1    | 0.8    | 0.1    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.5    | 2.9    | 1.2    | 0.9    | 0.1    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.5    | 2.8    | 1.2    | 0.9    | 0.1    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.5    | 2.7    | 1.1    | 0.8    | 0.1    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.5    | 2.7    | 1      | 0.8    | 0.1    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.4    | 2.6    | 0.9    | 0.8    | 0.1    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.5    | 2.6    | 0.9    | 0.8    | 0.1    |        |

# Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two\_stage - IPDE alpha = 0.6 - both utilities - N = 30

|                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|--------------------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 27                          |        |        |        |        |        |        |
| DLT rate                             | 0.11   | 0.17   | 0.3    | 0.5    | 0.62   |        |
| Efficacy rate                        | 0.1    | 0.2    | 0.4    | 0.42   | 0.45   |        |
| Utility 2                            | 0      | 0      | 0.1    | -0.1   | -0.2   |        |
| Utility 3                            | 41.6   | 45.2   | 52     | 45.2   | 42.2   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 22.7   | 26.1   | 38.6   | 6.5    | 0.7    | 5.4    |
| U3 - Toxicity prior Beta(0.15, 0.85) | 27.5   | 25.9   | 34.5   | 6.1    | 0.6    | 5.4    |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 22.3   | 24.7   | 38.2   | 8.5    | 0.8    | 5.5    |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 25.5   | 26.5   | 34.5   | 7.5    | 0.5    | 5.5    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 22.8   | 22.9   | 39.1   | 9      | 0.9    | 5.3    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 26.7   | 24.2   | 35.8   | 7      | 1      | 5.3    |
| U2 - Toxicity prior Beta(1, 1)       | 22.7   | 22.3   | 38.9   | 9.5    | 1.2    | 5.4    |
| U3 - Toxicity prior Beta(1, 1)       | 26.9   | 23.9   | 34.1   | 8.8    | 0.9    | 5.4    |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 8.2    | 6.9    | 9      | 3.9    | 1.4    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 8.2    | 6.9    | 9.1    | 3.8    | 1.3    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 7.9    | 6.2    | 9      | 4.8    | 1.5    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 7.9    | 6.4    | 8.8    | 4.8    | 1.6    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 7.8    | 6      | 8.8    | 5.2    | 1.7    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 7.7    | 6.1    | 8.8    | 5.2    | 1.7    |        |
| U2 - Toxicity prior Beta(1, 1)       | 7.7    | 5.9    | 8.5    | 5.4    | 1.9    |        |
| U3 - Toxicity prior Beta(1, 1)       | 7.7    | 6.1    | 8.5    | 5.3    | 1.9    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.5    | 3      | 1.6    | 1      | 0.2    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.5    | 3      | 1.6    | 1      | 0.2    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.5    | 2.8    | 1.6    | 1.1    | 0.2    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.5    | 2.8    | 1.6    | 1.1    | 0.2    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.4    | 2.8    | 1.5    | 1.1    | 0.2    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.4    | 2.8    | 1.5    | 1.1    | 0.2    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.4    | 2.7    | 1.3    | 1      | 0.2    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.4    | 2.7    | 1.3    | 1      | 0.2    |        |
| Scenario 28                          |        |        |        |        |        |        |
| DLT rate                             | 0.2    | 0.3    | 0.5    | 0.6    | 0.77   |        |
| Efficacy rate                        | 0.05   | 0.3    | 0.32   | 0.35   | 0.4    |        |
| Utility 2                            | -0.1   | 0      | -0.2   | -0.2   | -0.4   |        |
| Utility 3                            | 35     | 46     | 39.2   | 37     | 33.2   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 25.2   | 35.6   | 6.1    | 0      | 0      | 33.1   |
| U3 - Toxicity prior Beta(0.15, 0.85) | 25.6   | 35.5   | 5.5    | 0.1    | 0      | 33.3   |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 24.9   | 36.6   | 6.1    | 0.2    | 0      | 32.2   |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 25.8   | 36.5   | 4.8    | 0.2    | 0      | 32.7   |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 26.4   | 33.5   | 7.4    | 0.4    | 0      | 32.3   |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 26.8   | 35.7   | 4.9    | 0.4    | 0      | 32.2   |
| U2 - Toxicity prior Beta(1, 1)       | 26.6   | 33.8   | 6.7    | 0.7    | 0      | 32.2   |
| U3 - Toxicity prior Beta(1, 1)       | 28.3   | 33.1   | 5.5    | 0.6    | 0      | 32.5   |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 12.4   | 8.6    | 3.9    | 0.9    | 0.2    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 12.4   | 8.7    | 3.9    | 0.9    | 0.2    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 11.9   | 8.7    | 4.3    | 1.2    | 0.2    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 11.9   | 8.6    | 4.3    | 1.2    | 0.2    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 11.7   | 8.2    | 4.7    | 1.5    | 0.2    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 11.7   | 8.2    | 4.7    | 1.5    | 0.2    |        |
| U2 - Toxicity prior Beta(1, 1)       | 11.5   | 8.4    | 4.6    | 1.6    | 0.3    |        |
| U3 - Toxicity prior Beta(1, 1)       | 11.5   | 8.4    | 4.7    | 1.6    | 0.3    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.4    | 2.8    | 0.7    | 0.3    | 0      |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.4    | 2.8    | 0.7    | 0.3    | 0      |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.4    | 2.6    | 0.7    | 0.3    | 0      |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.4    | 2.6    | 0.7    | 0.3    | 0      |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.3    | 2.5    | 0.7    | 0.3    | 0      |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.3    | 2.5    | 0.7    | 0.3    | 0      |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.2    | 2.2    | 0.6    | 0.3    | 0      |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.2    | 2.2    | 0.6    | 0.3    | 0      |        |

# Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two\_stage - IPDE alpha = 0.6 - both utilities - N = 30

|                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|--------------------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 29                          |        |        |        |        |        |        |
| DLT rate                             | 0.19   | 0.3    | 0.56   | 0.74   | 0.86   |        |
| Efficacy rate                        | 0.4    | 0.4    | 0.4    | 0.4    | 0.4    |        |
| Utility 2                            | 0.2    | 0.1    | -0.2   | -0.3   | -0.5   |        |
| Utility 3                            | 56.4   | 52     | 41.6   | 34.4   | 29.6   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 69.5   | 24.2   | 3.4    | 0.3    | 0      | 2.6    |
| U3 - Toxicity prior Beta(0.15, 0.85) | 72.4   | 21.1   | 3.3    | 0.5    | 0      | 2.7    |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 68.8   | 24.5   | 3.5    | 0.4    | 0      | 2.8    |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 71.7   | 22     | 3.5    | 0.1    | 0      | 2.7    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 69.3   | 24.2   | 3.7    | 0.2    | 0      | 2.6    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 70.6   | 23.7   | 2.8    | 0.2    | 0      | 2.7    |
| U2 - Toxicity prior Beta(1, 1)       | 69.1   | 24.6   | 3.6    | 0.3    | 0      | 2.4    |
| U3 - Toxicity prior Beta(1, 1)       | 71.1   | 22.5   | 3.5    | 0.3    | 0      | 2.6    |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 18     | 7.6    | 3.3    | 0.6    | 0      |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 18.2   | 7.5    | 3.3    | 0.5    | 0      |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 17.4   | 7.7    | 3.5    | 0.8    | 0      |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 17.5   | 7.6    | 3.5    | 0.8    | 0      |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 17     | 7.9    | 3.7    | 0.9    | 0.1    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 17.1   | 7.8    | 3.7    | 0.9    | 0.1    |        |
| U2 - Toxicity prior Beta(1, 1)       | 16.9   | 8      | 3.7    | 0.9    | 0.1    |        |
| U3 - Toxicity prior Beta(1, 1)       | 17     | 7.9    | 3.7    | 0.9    | 0.1    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.7    | 2      | 0.7    | 0.1    | 0      |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.7    | 2      | 0.7    | 0.1    | 0      |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.7    | 1.9    | 0.7    | 0.2    | 0      |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.7    | 1.9    | 0.7    | 0.2    | 0      |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.6    | 1.8    | 0.6    | 0.2    | 0      |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.6    | 1.8    | 0.6    | 0.2    | 0      |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.6    | 1.7    | 0.6    | 0.1    | 0      |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.6    | 1.7    | 0.6    | 0.1    | 0      |        |
| Scenario 30                          |        |        |        |        |        |        |
| DLT rate                             | 0.05   | 0.08   | 0.17   | 0.27   | 0.54   |        |
| Efficacy rate                        | 0.12   | 0.15   | 0.2    | 0.4    | 0.5    |        |
| Utility 2                            | 0.1    | 0.1    | 0      | 0.1    | 0      |        |
| Utility 3                            | 45.2   | 45.8   | 45.2   | 53.2   | 48.4   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 19.1   | 17.9   | 17.5   | 34.5   | 10.2   | 0.8    |
| U3 - Toxicity prior Beta(0.15, 0.85) | 28.3   | 21.4   | 13.4   | 28.5   | 7.6    | 0.8    |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 19.2   | 16     | 14.8   | 39.2   | 10     | 0.8    |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 25.6   | 19.3   | 12.6   | 34.3   | 7.4    | 0.8    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 18.3   | 17.3   | 14.7   | 37.3   | 11.6   | 0.8    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 25.6   | 21     | 11.9   | 31.8   | 8.9    | 0.8    |
| U2 - Toxicity prior Beta(1, 1)       | 20.4   | 14.6   | 13.1   | 37.8   | 13.3   | 0.8    |
| U3 - Toxicity prior Beta(1, 1)       | 26.7   | 18.5   | 12.6   | 31.1   | 10.3   | 0.8    |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 5.1    | 4.8    | 6      | 8      | 6      |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 5.2    | 4.7    | 6      | 8.1    | 5.9    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 4.9    | 4.5    | 5.8    | 8.5    | 6.3    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 4.9    | 4.4    | 5.7    | 8.5    | 6.3    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 4.8    | 4.4    | 5.5    | 8.6    | 6.6    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 4.9    | 4.3    | 5.6    | 8.5    | 6.6    |        |
| U2 - Toxicity prior Beta(1, 1)       | 4.8    | 4.3    | 5.4    | 8.4    | 7.1    |        |
| U3 - Toxicity prior Beta(1, 1)       | 4.8    | 4.3    | 5.4    | 8.4    | 7      |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.4    | 2.8    | 1.6    | 2.3    | 0.9    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.4    | 2.8    | 1.6    | 2.2    | 0.9    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.3    | 2.7    | 1.5    | 2.3    | 0.9    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.4    | 2.7    | 1.6    | 2.3    | 0.9    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.3    | 2.7    | 1.5    | 2.2    | 0.9    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.3    | 2.7    | 1.5    | 2.2    | 0.9    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.3    | 2.7    | 1.4    | 2.1    | 0.8    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.4    | 2.7    | 1.4    | 2.1    | 0.8    |        |

# Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two\_stage - IPDE alpha = 0.6 - both utilities - N = 30

|                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|--------------------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 31                          |        |        |        |        |        |        |
| DLT rate                             | 0.04   | 0.06   | 0.1    | 0.2    | 0.6    |        |
| Efficacy rate                        | 0.2    | 0.5    | 0.5    | 0.5    | 0.5    |        |
| Utility 2                            | 0.2    | 0.4    | 0.4    | 0.3    | -0.1   |        |
| Utility 3                            | 50.4   | 67.6   | 66     | 62     | 46     |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 7.2    | 31.5   | 28.3   | 25.3   | 7.6    | 0.1    |
| U3 - Toxicity prior Beta(0.15, 0.85) | 8.9    | 37.5   | 28.7   | 20     | 4.8    | 0.1    |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 6.4    | 31.7   | 28.2   | 26.1   | 7.5    | 0.1    |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 8.3    | 38.1   | 26.7   | 21.5   | 5.3    | 0.1    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 6.2    | 33.4   | 27.1   | 26.3   | 6.9    | 0.1    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 8.6    | 38.7   | 26.2   | 22.7   | 3.7    | 0.1    |
| U2 - Toxicity prior Beta(1, 1)       | 6      | 33     | 27.1   | 27.6   | 6.2    | 0.1    |
| U3 - Toxicity prior Beta(1, 1)       | 8.7    | 38.9   | 26.4   | 22     | 3.9    | 0.1    |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 4.1    | 5.6    | 6.5    | 8      | 5.8    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 4.1    | 5.6    | 6.5    | 7.9    | 5.9    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 4      | 5.5    | 6.3    | 8.1    | 6.1    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 4      | 5.5    | 6.4    | 8      | 6.1    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 4      | 5.4    | 6.3    | 8.1    | 6.2    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 4      | 5.4    | 6.3    | 8      | 6.2    |        |
| U2 - Toxicity prior Beta(1, 1)       | 4      | 5.4    | 6.2    | 8      | 6.4    |        |
| U3 - Toxicity prior Beta(1, 1)       | 4      | 5.4    | 6.2    | 8      | 6.4    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.1    | 2.8    | 1.4    | 1.8    | 1.1    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.1    | 2.8    | 1.4    | 1.8    | 1.1    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.1    | 2.7    | 1.3    | 1.8    | 1.1    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.1    | 2.7    | 1.4    | 1.8    | 1.1    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.1    | 2.7    | 1.3    | 1.8    | 1.1    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.1    | 2.7    | 1.3    | 1.8    | 1.1    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.2    | 2.7    | 1.3    | 1.8    | 1.1    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.1    | 2.7    | 1.3    | 1.8    | 1      |        |
| Scenario 32                          |        |        |        |        |        |        |
| DLT rate                             | 0.02   | 0.05   | 0.11   | 0.3    | 0.36   |        |
| Efficacy rate                        | 0.08   | 0.15   | 0.3    | 0.3    | 0.3    |        |
| Utility 2                            | 0.1    | 0.1    | 0.2    | 0      | -0.1   |        |
| Utility 3                            | 44     | 47     | 53.6   | 46     | 43.6   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 10.6   | 15.7   | 32.4   | 29.2   | 11.5   | 0.6    |
| U3 - Toxicity prior Beta(0.15, 0.85) | 16.4   | 19.7   | 31.7   | 24.5   | 7.1    | 0.6    |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 10.7   | 14.2   | 32.9   | 28.9   | 12.7   | 0.6    |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 16.3   | 21.2   | 31.3   | 23.1   | 7.5    | 0.6    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 10     | 14.5   | 33.4   | 28.6   | 12.9   | 0.6    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 15.6   | 20.4   | 31.8   | 23.7   | 7.9    | 0.6    |
| U2 - Toxicity prior Beta(1, 1)       | 10.3   | 14.5   | 33.2   | 28     | 13.4   | 0.6    |
| U3 - Toxicity prior Beta(1, 1)       | 15.2   | 19.9   | 30.6   | 24.7   | 9      | 0.6    |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 4      | 4.3    | 7.5    | 7.4    | 6.8    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 4      | 4.3    | 7.6    | 7.4    | 6.7    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 3.9    | 4.2    | 7.4    | 7.5    | 7      |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 3.9    | 4.2    | 7.3    | 7.5    | 7      |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 3.9    | 4.1    | 7.2    | 7.5    | 7.2    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 4      | 4.1    | 7.3    | 7.4    | 7.2    |        |
| U2 - Toxicity prior Beta(1, 1)       | 3.9    | 4.1    | 7.1    | 7.5    | 7.3    |        |
| U3 - Toxicity prior Beta(1, 1)       | 3.9    | 4.1    | 7.1    | 7.5    | 7.3    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.3    | 2.8    | 1.9    | 2.7    | 1.2    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.3    | 2.8    | 1.9    | 2.7    | 1.2    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.3    | 2.8    | 1.9    | 2.6    | 1.2    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.3    | 2.8    | 1.9    | 2.7    | 1.2    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.3    | 2.8    | 1.9    | 2.6    | 1.2    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.3    | 2.8    | 1.9    | 2.6    | 1.2    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.3    | 2.8    | 1.8    | 2.6    | 1.1    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.3    | 2.8    | 1.9    | 2.6    | 1.2    |        |



# Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two\_stage - IPDE alpha = 0.6 - both utilities - N = 30

|                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|--------------------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 33                          |        |        |        |        |        |        |
| DLT rate                             | 0.01   | 0.02   | 0.09   | 0.32   | 0.38   |        |
| Efficacy rate                        | 0.05   | 0.08   | 0.2    | 0.4    | 0.5    |        |
| Utility 2                            | 0      | 0.1    | 0.1    | 0.1    | 0.1    |        |
| Utility 3                            | 42.6   | 44     | 48.4   | 51.2   | 54.8   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 9.6    | 6.1    | 14     | 40.7   | 29.6   | 0      |
| U3 - Toxicity prior Beta(0.15, 0.85) | 14.3   | 8.5    | 13.7   | 38.2   | 25.3   | 0      |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 8.6    | 5.5    | 12.7   | 41.7   | 31.5   | 0      |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 13     | 7.1    | 13.9   | 39.6   | 26.4   | 0      |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 8.9    | 4.6    | 13.1   | 41.4   | 32     | 0      |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 12.3   | 6.9    | 13.1   | 39.5   | 28.2   | 0      |
| U2 - Toxicity prior Beta(1, 1)       | 8.3    | 4.3    | 13.1   | 39.5   | 34.8   | 0      |
| U3 - Toxicity prior Beta(1, 1)       | 12.5   | 7.1    | 12.1   | 37.9   | 30.4   | 0      |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 3.6    | 3.6    | 5.2    | 8.3    | 9.4    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 3.5    | 3.5    | 5.2    | 8.3    | 9.4    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 3.5    | 3.5    | 5.1    | 8.2    | 9.7    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 3.5    | 3.5    | 5.1    | 8.3    | 9.7    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 3.5    | 3.5    | 5      | 8.2    | 9.9    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 3.5    | 3.5    | 4.9    | 8.3    | 9.8    |        |
| U2 - Toxicity prior Beta(1, 1)       | 3.5    | 3.4    | 4.8    | 8.2    | 10.1   |        |
| U3 - Toxicity prior Beta(1, 1)       | 3.4    | 3.5    | 4.8    | 8.2    | 10.1   |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.2    | 2.6    | 1.4    | 2.8    | 1      |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.2    | 2.6    | 1.4    | 2.8    | 1      |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.2    | 2.6    | 1.4    | 2.7    | 1      |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.2    | 2.6    | 1.4    | 2.7    | 1      |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.2    | 2.6    | 1.4    | 2.7    | 1      |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.2    | 2.6    | 1.4    | 2.7    | 1      |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.2    | 2.6    | 1.4    | 2.7    | 0.9    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.2    | 2.6    | 1.4    | 2.7    | 0.9    |        |
| Scenario 34                          |        |        |        |        |        |        |
| DLT rate                             | 0.05   | 0.13   | 0.3    | 0.36   | 0.42   |        |
| Efficacy rate                        | 0.05   | 0.1    | 0.3    | 0.32   | 0.35   |        |
| Utility 2                            | 0      | 0      | 0      | 0      | -0.1   |        |
| Utility 3                            | 41     | 40.8   | 46     | 44.8   | 44.2   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 17.3   | 15.7   | 44.3   | 16.4   | 4.5    | 1.8    |
| U3 - Toxicity prior Beta(0.15, 0.85) | 23.4   | 16.2   | 40.4   | 15.4   | 2.7    | 1.9    |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 16.6   | 15.5   | 41.5   | 19     | 5.6    | 1.8    |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 23.4   | 16.7   | 38     | 16.4   | 3.7    | 1.8    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 16     | 13.6   | 40.9   | 20.6   | 7.1    | 1.8    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 23.5   | 14.5   | 38.3   | 17.3   | 4.6    | 1.8    |
| U2 - Toxicity prior Beta(1, 1)       | 15.2   | 13     | 40.3   | 21.5   | 8.1    | 1.9    |
| U3 - Toxicity prior Beta(1, 1)       | 21.4   | 15.4   | 37.3   | 19     | 5.1    | 1.8    |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 5.4    | 6.1    | 9.4    | 5.5    | 3.3    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 5.4    | 6      | 9.4    | 5.6    | 3.3    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 5.1    | 5.7    | 8.8    | 6.4    | 3.7    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 5.2    | 5.6    | 9      | 6.3    | 3.7    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 5      | 5.4    | 8.8    | 6.6    | 4      |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 5      | 5.4    | 8.8    | 6.6    | 4      |        |
| U2 - Toxicity prior Beta(1, 1)       | 4.9    | 5.1    | 8.6    | 6.6    | 4.5    |        |
| U3 - Toxicity prior Beta(1, 1)       | 4.9    | 5.2    | 8.5    | 6.7    | 4.5    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.6    | 3.2    | 1.9    | 1.6    | 0.6    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.6    | 3.2    | 1.9    | 1.7    | 0.6    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.5    | 3.1    | 1.9    | 1.8    | 0.7    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.5    | 3.1    | 1.9    | 1.7    | 0.7    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.5    | 3      | 1.8    | 1.7    | 0.7    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.5    | 3      | 1.8    | 1.7    | 0.7    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.4    | 2.9    | 1.7    | 1.7    | 0.6    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.4    | 2.9    | 1.7    | 1.6    | 0.6    |        |

# Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two\_stage - IPDE alpha = 0.6 - both utilities - N = 30

|                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|--------------------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 35                          |        |        |        |        |        |        |
| DLT rate                             | 0.03   | 0.09   | 0.3    | 0.4    | 0.45   |        |
| Efficacy rate                        | 0.2    | 0.45   | 0.45   | 0.45   | 0.45   |        |
| Utility 2                            | 0.2    | 0.4    | 0.1    | 0.1    | 0      |        |
| Utility 3                            | 50.8   | 63.4   | 55     | 51     | 49     |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 12.1   | 40.3   | 33.8   | 11.2   | 2.5    | 0.1    |
| U3 - Toxicity prior Beta(0.15, 0.85) | 15.3   | 44.6   | 29.4   | 8.9    | 1.7    | 0.1    |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 11.4   | 40.5   | 32.1   | 13.1   | 2.8    | 0.1    |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 15     | 44.5   | 27.4   | 10.9   | 2.1    | 0.1    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 12.2   | 38.4   | 31.5   | 14.1   | 3.7    | 0.1    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 15.4   | 44.4   | 26.8   | 10.8   | 2.5    | 0.1    |
| U2 - Toxicity prior Beta(1, 1)       | 11.5   | 37.9   | 32.9   | 13.7   | 3.9    | 0.1    |
| U3 - Toxicity prior Beta(1, 1)       | 15.9   | 42.8   | 27.4   | 11.4   | 2.4    | 0.1    |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 4.6    | 8.3    | 9      | 5.3    | 2.9    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 4.6    | 8.2    | 9      | 5.3    | 2.8    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 4.5    | 8      | 8.5    | 5.9    | 3.1    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 4.5    | 8      | 8.6    | 5.8    | 3.1    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 4.5    | 7.7    | 8.4    | 6      | 3.4    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 4.4    | 7.7    | 8.4    | 6      | 3.4    |        |
| U2 - Toxicity prior Beta(1, 1)       | 4.4    | 7.7    | 8.1    | 6.1    | 3.7    |        |
| U3 - Toxicity prior Beta(1, 1)       | 4.4    | 7.7    | 8.2    | 6.1    | 3.6    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.3    | 3      | 1.6    | 1.4    | 0.5    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.3    | 3      | 1.6    | 1.4    | 0.5    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.3    | 3      | 1.6    | 1.4    | 0.5    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.3    | 3      | 1.6    | 1.4    | 0.5    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.3    | 3      | 1.5    | 1.4    | 0.5    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.3    | 3      | 1.6    | 1.4    | 0.5    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.3    | 3      | 1.5    | 1.4    | 0.5    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.3    | 3      | 1.5    | 1.3    | 0.5    |        |
| Scenario 36                          |        |        |        |        |        |        |
| DLT rate                             | 0.13   | 0.29   | 0.35   | 0.4    | 0.43   |        |
| Efficacy rate                        | 0.05   | 0.2    | 0.22   | 0.3    | 0.35   |        |
| Utility 2                            | -0.1   | -0.1   | -0.1   | -0.1   | -0.1   |        |
| Utility 3                            | 37.8   | 40.4   | 39.2   | 42     | 43.8   |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 24.2   | 40     | 12.9   | 5.7    | 2.1    | 15.1   |
| U3 - Toxicity prior Beta(0.15, 0.85) | 27.7   | 40.1   | 11.1   | 4.9    | 1.3    | 14.9   |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 22.6   | 38     | 14.9   | 6.8    | 2.9    | 14.8   |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 27.1   | 39.7   | 11.3   | 5.9    | 1.6    | 14.4   |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 22.6   | 37.3   | 14.7   | 7.1    | 3.9    | 14.4   |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 27.1   | 37.3   | 12.7   | 5.6    | 2.9    | 14.4   |
| U2 - Toxicity prior Beta(1, 1)       | 20.1   | 35.9   | 16.8   | 8.7    | 4      | 14.5   |
| U3 - Toxicity prior Beta(1, 1)       | 25.1   | 36.5   | 13.6   | 7.4    | 2.9    | 14.5   |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 9.5    | 8.6    | 5.4    | 3      | 1.7    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 9.6    | 8.7    | 5.3    | 3      | 1.7    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 8.8    | 8.2    | 5.9    | 3.5    | 2      |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 8.8    | 8.1    | 5.8    | 3.5    | 2      |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 8.6    | 7.8    | 5.9    | 3.9    | 2.2    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 8.6    | 7.7    | 6      | 3.9    | 2.1    |        |
| U2 - Toxicity prior Beta(1, 1)       | 8.5    | 7.7    | 5.7    | 4      | 2.4    |        |
| U3 - Toxicity prior Beta(1, 1)       | 8.5    | 7.8    | 5.7    | 4      | 2.4    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.6    | 3.4    | 1.1    | 0.9    | 0.2    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.5    | 3.5    | 1.1    | 0.9    | 0.2    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.5    | 3.3    | 1.1    | 1      | 0.2    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.4    | 3.2    | 1.2    | 1      | 0.2    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.4    | 3.1    | 1.1    | 1      | 0.2    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.4    | 3      | 1.1    | 1      | 0.2    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.4    | 2.9    | 0.9    | 0.9    | 0.2    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.4    | 2.9    | 0.9    | 0.9    | 0.2    |        |

# Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two\_stage - IPDE alpha = 0.6 - both utilities - N = 30

|                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|--------------------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 37                          |        |        |        |        |        |        |
| DLT rate                             | 0.08   | 0.3    | 0.4    | 0.45   | 0.5    |        |
| Efficacy rate                        | 0.05   | 0.2    | 0.2    | 0.2    | 0.2    |        |
| Utility 2                            | 0      | -0.1   | -0.2   | -0.2   | -0.3   |        |
| Utility 3                            | 39.8   | 40     | 36     | 34     | 32     |        |
| OBD selection (%)                    |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 30.6   | 46.6   | 12.9   | 2.5    | 0.5    | 6.9    |
| U3 - Toxicity prior Beta(0.15, 0.85) | 34.6   | 46.1   | 9.7    | 2.3    | 0.3    | 7      |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 28.7   | 45.7   | 14.4   | 4.1    | 0.7    | 6.4    |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 34.5   | 45.2   | 10.4   | 2.7    | 0.6    | 6.6    |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 29.5   | 44.8   | 14.7   | 3.7    | 1.1    | 6.2    |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 35.4   | 43.5   | 11.3   | 3.2    | 0.3    | 6.3    |
| U2 - Toxicity prior Beta(1, 1)       | 29.6   | 45     | 14.5   | 3.5    | 1.3    | 6.1    |
| U3 - Toxicity prior Beta(1, 1)       | 33.3   | 45.6   | 10.6   | 3.8    | 0.5    | 6.2    |
| Mean patients treated                |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 8.7    | 10.2   | 6.4    | 2.5    | 1.1    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 8.8    | 10.2   | 6.4    | 2.5    | 1.1    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 8.3    | 9.9    | 6.4    | 3.2    | 1.3    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 8.3    | 9.9    | 6.5    | 3.1    | 1.3    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 8.1    | 9.5    | 6.5    | 3.5    | 1.4    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 8      | 9.6    | 6.5    | 3.5    | 1.4    |        |
| U2 - Toxicity prior Beta(1, 1)       | 8      | 9.5    | 6.4    | 3.6    | 1.6    |        |
| U3 - Toxicity prior Beta(1, 1)       | 8      | 9.5    | 6.3    | 3.7    | 1.6    |        |
| Mean IPDE patients                   |        |        |        |        |        |        |
| U2 - Toxicity prior Beta(0.15, 0.85) | 0.7    | 3.9    | 1.3    | 0.9    | 0.2    |        |
| U3 - Toxicity prior Beta(0.15, 0.85) | 0.8    | 4      | 1.2    | 0.9    | 0.2    |        |
| U2 - Toxicity prior Beta(0.3, 0.7)   | 0.7    | 3.8    | 1.2    | 1      | 0.2    |        |
| U3 - Toxicity prior Beta(0.3, 0.7)   | 0.7    | 3.8    | 1.3    | 0.9    | 0.2    |        |
| U2 - Toxicity prior Beta(0.5, 0.5)   | 0.7    | 3.5    | 1.2    | 0.9    | 0.2    |        |
| U3 - Toxicity prior Beta(0.5, 0.5)   | 0.7    | 3.5    | 1.2    | 0.9    | 0.2    |        |
| U2 - Toxicity prior Beta(1, 1)       | 0.6    | 3.3    | 1      | 0.9    | 0.2    |        |
| U3 - Toxicity prior Beta(1, 1)       | 0.6    | 3.3    | 1      | 0.9    | 0.2    |        |